

Test Evidence and Bug Report

I built the program with `gcc -Wall -Wextra` and it compiled clean, no errors and no warnings. This release adds academic standing to SIMS, so the program now checks the GPA, works out whether the student is in Honors, Good Standing, Academic Probation or Academic Suspension, and prints that in the student report.

I ran nine tests. The first six are the required academic standing cases from the CCR, and the last three are the Part 2 regression tests, so I know the course grade work from last week still does what it did. All nine passed. The screenshots are on the pages after this one.

Bugs I ran into

The GPA on the report did not always agree with the standing

This is the one that actually bit me, and I only caught it because I tried a GPA with three decimals. The report prints the GPA with `%.2f`, which rounds it to two decimals, but the classification was reading the raw number I typed in. Those two do not always agree. When I entered 1.999 the program printed `Current GPA : 2.00` on one line and `Academic Standing : Academic Probation` on the next, which contradicts itself, because a 2.00 GPA is Good Standing. The display rounded up across the band edge and the classification did not follow it.

I fixed it by rounding the GPA to two decimals one time, into a variable I called `gpaRounded`, and then using that same variable for both the classification and the line that prints. It multiplies by 100, adds 0.5 so the cast rounds to nearest instead of chopping the rest off, casts to `int`, then divides by 100.0f. That is the same cast trick I already used for the average grade in Part 2. Since both the printed GPA and the classified GPA now read the same variable they cannot disagree. I re-ran every test after the fix and 1.999 now comes out as 2.00 with Good Standing, and none of the required cases or the Part 2 cases changed.

A stale copy of main.c that would have undone a Part 2 fix

Before I wrote any Part 3 code I found that my working `main.c` and the snapshot I saved in the Part 2 folder had drifted apart. Both were the same length, but the working copy still printed the Student Summary before the five course grade prompts, and the snapshot had the Part 2 fix that moved the summary down so the student information and the grade information come out together. If I had built Part 3 on the working copy I would have quietly undone that fix and failed a Part 2 acceptance criterion. I compared the two, confirmed the snapshot was the newer one, and copied it over the working file before starting.

Nothing else broke

Other than those two the build was clean. The Part 3 changes only add to the program, so the average, highest and lowest logic and the Part 1 input checks were not touched, and all three Part 2 regression cases print the same numbers they did last week.

Assumptions I made

The CCR asks me to write down any assumptions I made where the requirements were ambiguous. There were four.

1. The GPA bands have gaps in them, so I used lower bounds only

The policy table lists 3.50 to 4.00, 2.00 to 3.49, 1.00 to 1.99 and 0.00 to 0.99. Those ranges do not touch each other. A GPA of 3.495 sits between the top of Good Standing and the bottom of Honors, so reading the table literally it would not be classified as anything. I assumed the table was written expecting two decimal places rather than meaning to leave a hole, so I wrote the classification as a chain of lower bounds only, 3.50 then 2.00 then 1.00 then everything else. That closes the gaps, treats the top and bottom of each band as included, and means every valid GPA lands in exactly one category.

2. A GPA is a two decimal number

The CCR asks whether a GPA can have more than two decimal places but does not answer it, and every example in the CCR uses two. I assumed two decimals is the real format, so the program rounds the GPA to two decimals before it classifies or displays it. This is also what fixed the contradiction bug on the previous page.

3. I kept the Programmer line and the Welcome banner even though this CCR's example leaves them out

The expected output in this CCR does not show the Programmer line or the Welcome to SIMS banner, but the acceptance criteria still says all CCR-001 functionality has to keep working, and CCR-001 asked for a welcome banner. I assumed the example just carried an omission forward from editing the template, so I kept both. This is the same call I made and wrote up in Part 2.

4. An invalid GPA ends the program instead of asking again

The CCR asks whether the program should keep going after an invalid GPA and whether the user should get to re-enter it. Asking again needs a loop, and loops are not on this release's permitted list, so I kept what I did in Part 1 and Part 2: print the error and exit. Trying again means running the program again.

Answers to the Questions for the Customer

1. Are the GPA ranges inclusive of the upper and lower limits?

Yes, both ends are included. A GPA of exactly 3.50 is Honors, exactly 2.00 is Good Standing, exactly 1.00 is Academic Probation and exactly 0.00 is Academic Suspension. 4.00 is Honors. I checked each of those boundary values myself.

2. Can a GPA contain more than two decimal places?

The program will take it, but it treats the GPA as a two decimal number. It rounds to two decimals and then uses that for both the display and the classification, so 3.495 shows as 3.50 and comes out as Honors. This one is a judgment call, since the CCR never states a precision but every example uses two decimals.

3. Should the program continue processing after an invalid GPA is entered?

No. It prints the error and exits without asking for course grades and without working out a standing, which is what the CCR asks for. It also matches how Part 1 and Part 2 handle bad input.

4. Should the user be allowed to re-enter an invalid GPA?

Not in this release. Asking again needs a loop and loops are not on the permitted list this week, so the user runs the program again. If the Registrar wants a retry inside the program that would need a future CCR once loops are available.

5. What message should be displayed for an invalid GPA?

Exactly the three lines from Example 5 in the CCR: ERROR on its own line, then Invalid GPA entered., then GPA must be between 0.00 and 4.00. I matched that wording exactly.

6. Should the academic standing be stored or only displayed?

Stored first, then displayed. The result goes into an enum variable and the report reads from that. Storing it keeps the decision separate from the wording, so if the Registrar renamed a category later I would only change the text, not the logic.

7. Should the GPA always be entered manually, or will it eventually be calculated from course grades?

Manually for now. Automatic GPA calculation is listed as out of scope in this CCR. I assume it is coming later, since the program already collects the five course grades a calculation like that would need, so I kept GPA entry and grade entry independent instead of tying them together.

8. Should Honors students receive additional recognition in the report?

No extra recognition this release. The acceptance criteria only asks that the standing is worked out correctly and shown in the report, so Honors prints the same way the other three do.

9. Will future versions support different academic standing policies for graduate students?

Not addressed this release. There is one policy and it matches the table in this CCR. Supporting a second one would mean picking between rule sets while the program runs, which would need a future CCR.

10. Should academic standing be represented internally using descriptive text or predefined categories?

Predefined categories. I declared an enum with four constants instead of storing the text. I also needed it for the display, because a switch statement in C can only test an integer, so the float GPA gets reduced to one of the four enum values by the if/else chain first and the switch then picks the label.

11. Should the report include both the GPA and the academic standing?

Yes, both. Example 1 in the CCR shows Current GPA and Academic Standing as two lines next to each other in the Student Summary, so the standing prints right underneath the GPA in the same alignment.

12. Should academic standing be recalculated every time the GPA changes?

It does not come up this release. The program handles one student and reads the GPA once, and editing grades or GPA is out of scope, so there is no point where a stored GPA changes after it has been classified. Once editing exists the standing would have to be worked out again at that point.

Test 1: GPA 3.95, Honors

Input entered: Student ID 1234567, Student Name Alice Johnson, Current GPA 3.95, course grades 95, 88, 91, 84, 100.

Expected output: Academic Standing : Honors. Part 2 values unchanged: average 91.60, highest 100, lowest 84.

Actual output: Shown in the screenshot below.

Pass/Fail result: Pass

```
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Student Information Management System
Version 3.0

Programmer: Branson Zemaitis

Welcome to SIMS
-----

Enter Student ID: 1234567
Enter Student Name: Alice Johnson
Enter Current GPA: 3.95

Enter Course 1 Grade: 95
Enter Course 2 Grade: 88
Enter Course 3 Grade: 91
Enter Course 4 Grade: 84
Enter Course 5 Grade: 100

Student Summary

Student ID : 1234567
Student Name : Alice Johnson
Current GPA : 3.95
Academic Standing : Honors
-----

Course Grades

Course 1 : 95
Course 2 : 88
Course 3 : 91
Course 4 : 84
Course 5 : 100

-----

Average Grade : 91.60
Highest Grade : 100
Lowest Grade : 84
student@0193210b33c3:/workspace$
```

Test 2: GPA 3.20, Good Standing

Input entered: Student ID 1002003, Student Name Michael Brown, Current GPA 3.20, course grades 75, 82, 91, 68, 79.

Expected output: Academic Standing : Good Standing. Part 2 values: average 79.00, highest 91, lowest 68.

Actual output: Shown in the screenshot below.

Pass/Fail result: Pass

```
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Student Information Management System
Version 3.0

Programmer: Branson Zemaitis

Welcome to SIMS
-----

Enter Student ID: 1002003
Enter Student Name: Michael Brown
Enter Current GPA: 3.20

Enter Course 1 Grade: 75
Enter Course 2 Grade: 82
Enter Course 3 Grade: 91
Enter Course 4 Grade: 68
Enter Course 5 Grade: 79

Student Summary

Student ID : 1002003
Student Name : Michael Brown
Current GPA : 3.20
Academic Standing : Good Standing
-----
Course Grades

Course 1 : 75
Course 2 : 82
Course 3 : 91
Course 4 : 68
Course 5 : 79

-----
Average Grade : 79.00
Highest Grade : 91
Lowest Grade : 68
student@0193210b33c3: /workspace$
```

Test 3: GPA 1.75, Academic Probation

Input entered: Student ID 1122334, Student Name Jordan Lee, Current GPA 1.75, course grades 70, 65, 80, 72, 68.

Expected output: Academic Standing : Academic Probation. Part 2 values: average 71.00, highest 80, lowest 65.

Actual output: Shown in the screenshot below.

Pass/Fail result: Pass

```
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Student Information Management System
Version 3.0

Programmer: Branson Zemaitis

Welcome to SIMS
-----

Enter Student ID: 1122334
Enter Student Name: Jordan Lee
Enter Current GPA: 1.75

Enter Course 1 Grade: 70
Enter Course 2 Grade: 65
Enter Course 3 Grade: 80
Enter Course 4 Grade: 72
Enter Course 5 Grade: 68

Student Summary

Student ID : 1122334
Student Name : Jordan Lee
Current GPA : 1.75
Academic Standing : Academic Probation
-----

Course Grades

Course 1 : 70
Course 2 : 65
Course 3 : 80
Course 4 : 72
Course 5 : 68

-----

Average Grade : 71.00
Highest Grade : 80
Lowest Grade : 65
student@0193210b33c3:/workspace$
```

Test 4: GPA 0.60, Academic Suspension

Input entered: Student ID 2233445, Student Name Casey Morgan, Current GPA 0.60, course grades 55, 60, 48, 52, 61.

Expected output: Academic Standing : Academic Suspension. Part 2 values: average 55.20, highest 61, lowest 48.

Actual output: Shown in the screenshot below.

Pass/Fail result: Pass

```
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Student Information Management System  
Version 3.0  
  
Programmer: Branson Zemaitis  
  
Welcome to SIMS  
-----  
  
Enter Student ID: 2233445  
Enter Student Name: Casey Morgan  
Enter Current GPA: 0.60  
  
Enter Course 1 Grade: 55  
Enter Course 2 Grade: 60  
Enter Course 3 Grade: 48  
Enter Course 4 Grade: 52  
Enter Course 5 Grade: 61  
  
Student Summary  
  
Student ID : 2233445  
Student Name : Casey Morgan  
Current GPA : 0.60  
Academic Standing : Academic Suspension  
-----  
Course Grades  
  
Course 1 : 55  
Course 2 : 60  
Course 3 : 48  
Course 4 : 52  
Course 5 : 61  
  
-----  
Average Grade : 55.20  
Highest Grade : 61  
Lowest Grade : 48  
❖ student@0193210b33c3: /workspace$
```

Test 5: GPA -0.50, invalid

Input entered: Student ID 3344556, Student Name Riley Parker, Current GPA -0.50.

Expected output: ERROR / Invalid GPA entered. / GPA must be between 0.00 and 4.00. No academic standing determined, and no course grade prompts.

Actual output: Shown in the screenshot below.

Pass/Fail result: Pass

```
-----  
Student Information Management System  
Version 3.0  
  
Programmer: Branson Zemaitis  
  
Welcome to SIMS  
-----  
  
Enter Student ID: 3344556  
Enter Student Name: Riley Parker  
Enter Current GPA: -0.50  
ERROR  
Invalid GPA entered.  
GPA must be between 0.00 and 4.00.  
❖ student@0193210b33c3: /workspace$ █
```

Test 6: GPA 4.25, invalid

Input entered: Student ID 4455667, Student Name Taylor Quinn, Current GPA 4.25.

Expected output: ERROR / Invalid GPA entered. / GPA must be between 0.00 and 4.00. No academic standing determined, and no course grade prompts.

Actual output: Shown in the screenshot below.

Pass/Fail result: Pass

```
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Student Information Management System  
Version 3.0  
  
Programmer: Branson Zemaitis  
  
Welcome to SIMS  
-----  
  
Enter Student ID: 4455667  
Enter Student Name: Taylor Quinn  
Enter Current GPA: 4.25  
ERROR  
Invalid GPA entered.  
GPA must be between 0.00 and 4.00.  
❖ student@0193210b33c3:/workspace$
```

Test 7: CCR-002 regression, Alice Johnson

Input entered: Student ID 1234567, Student Name Alice Johnson, Current GPA 3.84, course grades 95, 88, 91, 84, 100.

Expected output: Average 91.60, highest 100, lowest 84, with Academic Standing : Honors added.

Actual output: Shown in the screenshot below.

Pass/Fail result: Pass

```
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Student Information Management System
Version 3.0

Programmer: Branson Zemaitis

Welcome to SIMS
-----

Enter Student ID: 1234567
Enter Student Name: Alice Johnson
Enter Current GPA: 3.84

Enter Course 1 Grade: 95
Enter Course 2 Grade: 88
Enter Course 3 Grade: 91
Enter Course 4 Grade: 84
Enter Course 5 Grade: 100

Student Summary

Student ID : 1234567
Student Name : Alice Johnson
Current GPA : 3.84
Academic Standing : Honors
-----

Course Grades

Course 1 : 95
Course 2 : 88
Course 3 : 91
Course 4 : 84
Course 5 : 100

-----

Average Grade : 91.60
Highest Grade : 100
Lowest Grade : 84
student@0193210b33c3:/workspace$ █
```

Test 8: CCR-002 regression, Michael Brown

Input entered: Student ID 1002003, Student Name Michael Brown, Current GPA 2.91, course grades 75, 82, 91, 68, 79.

Expected output: Average 79.00, highest 91, lowest 68, with Academic Standing : Good Standing.

Actual output: Shown in the screenshot below.

Pass/Fail result: Pass

```
-----  
Student Information Management System  
Version 3.0  
  
Programmer: Branson Zemaitis  
  
Welcome to SIMS  
-----  
  
Enter Student ID: 1002003  
Enter Student Name: Michael Brown  
Enter Current GPA: 2.91  
  
Enter Course 1 Grade: 75  
Enter Course 2 Grade: 82  
Enter Course 3 Grade: 91  
Enter Course 4 Grade: 68  
Enter Course 5 Grade: 79  
  
Student Summary  
  
Student ID : 1002003  
Student Name : Michael Brown  
Current GPA : 2.91  
Academic Standing : Good Standing  
-----  
Course Grades  
  
Course 1 : 75  
Course 2 : 82  
Course 3 : 91  
Course 4 : 68  
Course 5 : 79  
  
-----  
Average Grade : 79.00  
Highest Grade : 91  
Lowest Grade : 68  
student@0193210b33c3: /workspace$
```

Test 9: CCR-002 regression, Christopher Williams

Input entered: Student ID 9876543, Student Name Christopher Williams, Current GPA 4.00, course grades 100, 100, 100, 100, 100.

Expected output: Average 100.00, highest 100, lowest 100, with Academic Standing : Honors. Also covers the tie case, since every grade is the same the highest and the lowest are both 100.

Actual output: Shown in the screenshot below.

Pass/Fail result: Pass

```
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Student Information Management System
Version 3.0

Programmer: Branson Zemaitis

Welcome to SIMS
-----

Enter Student ID: 9876543
Enter Student Name: Christopher Williams
Enter Current GPA: 4.00

Enter Course 1 Grade: 100
Enter Course 2 Grade: 100
Enter Course 3 Grade: 100
Enter Course 4 Grade: 100
Enter Course 5 Grade: 100

Student Summary

Student ID : 9876543
Student Name : Christopher Williams
Current GPA : 4.00
Academic Standing : Honors
-----

Course Grades

Course 1 : 100
Course 2 : 100
Course 3 : 100
Course 4 : 100
Course 5 : 100

-----

Average Grade : 100.00
Highest Grade : 100
Lowest Grade : 100
student@0193210b33c3: /workspace$
```